

Breast Tumor Classification with Random Forest + Pattern Segmentation

UCI Breast Cancer Wisconsin (Diagnostic) Dataset

Student	Gokavarapu Syam Naga Sai
Project Type	Individual
Dataset	UCI Breast Cancer Wisconsin (Diagnostic)
Supervised Model	Random Forest Classifier (single algorithm)
Purpose	Learning / demonstration only; not a clinical diagnostic tool

1. Executive Summary

The project classifies tumors as malignant (M) or benign (B) using 30 quantitative diagnostic features. The dataset contains 569 observations: 357 benign and 212 malignant. A stratified 70/15/15 split with random seed 42 was used. Randomized search sampled the required Random Forest tuning space; the selected model used `{'n_estimators': 800, 'min_samples_split': 2, 'min_samples_leaf': 4, 'max_features': 'sqrt', 'max_depth': None, 'class_weight': None}`.

Final test results at threshold 0.20: ROC-AUC 0.999, malignant recall 1.000, precision 0.914, F1 0.955, accuracy 0.965.

2. Problem Definition and Success Metrics

Diagnosis was encoded M=1 and B=0. Primary metrics were ROC-AUC, malignant recall and F1. False negatives are malignant cases predicted as benign; false positives are benign cases predicted as malignant. The threshold was selected on validation to emphasize malignant recall.

3. Dataset Overview and Quality Checks

The supplied data contain 569 rows, 30 real-valued features, no missing values, zero duplicate rows and zero duplicate IDs. The ID field was excluded from training. Class distribution is B=357 (62.7%) and M=212 (37.3%).

4. Data Preparation and Split Strategy

The data were stratified into 397 training, 86 validation and 86 test observations. Random seed: 42. The test set was reserved for final reporting.

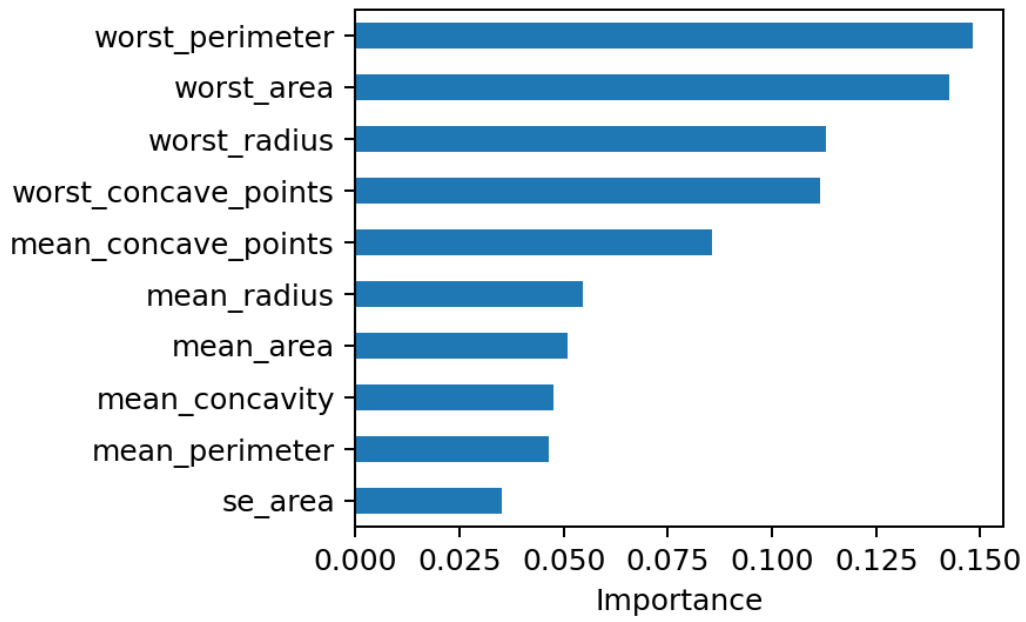
5. Feature Handling

All 30 features were retained for Random Forest. StandardScaler was used for clustering only.

6. Model Development and Tuning

RandomizedSearchCV with 3-fold stratified CV sampled six configurations from the required grid. Best CV ROC-AUC: 0.985. Validation ROC-AUC: 0.994. Final parameters: `{'n_estimators': 800, 'min_samples_split': 2, 'min_samples_leaf': 4, 'max_features': 'sqrt', 'max_depth': None, 'class_weight': None}`.

Top 10 Random Forest Features



7. Final Evaluation — Test Set

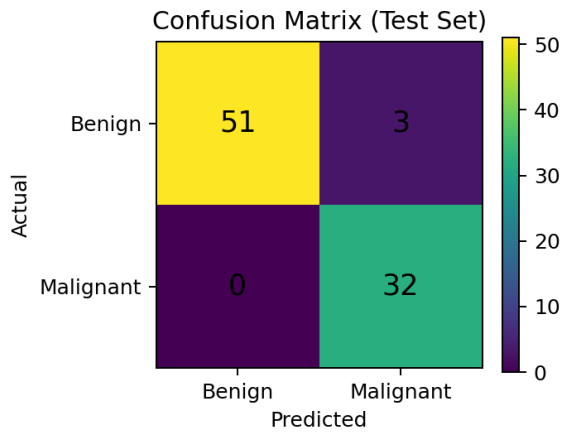
ACCURACY: 0.965

PRECISION: 0.914

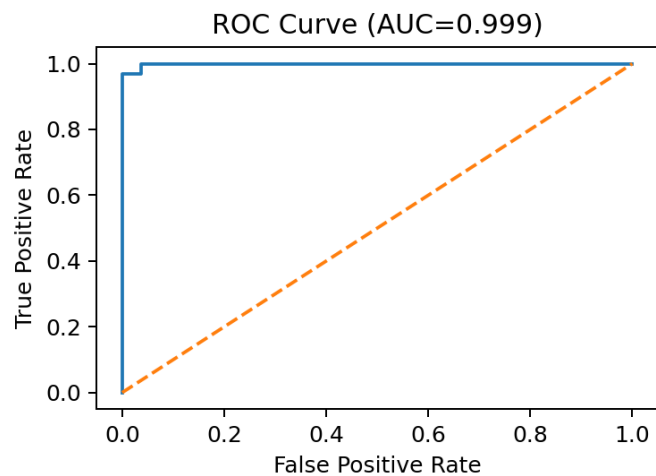
RECALL: 1.000

F1: 0.955

AUC: 0.999



Test confusion matrix: 51 true negatives, 3 false positives, 0 false negatives and 32 true positives.



Threshold 0.20 was fixed from validation; it produced 1.000 malignant recall on validation and 1.000 malignant recall on the test set. This is a project threshold, not a clinical threshold.

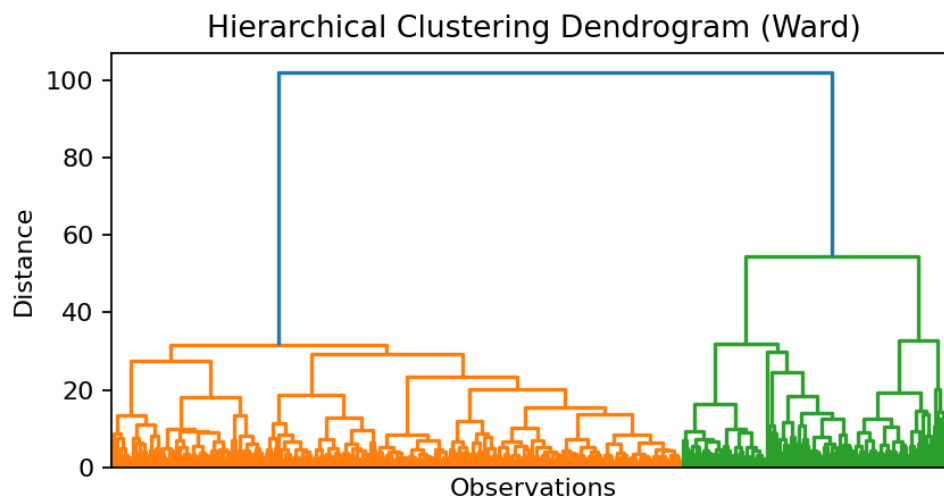
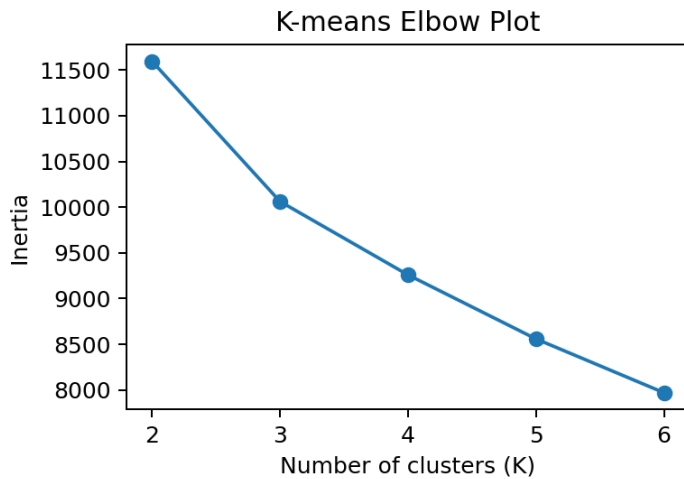
8. Error Analysis

There were no false negatives at the selected threshold in the test sample, so no observed FN pattern can be summarized. The three false positives had predicted malignancy probabilities of approximately 0.290, 0.267 and 0.227 and showed elevated compactness, fractal-dimension, concave-point and smoothness-related measures relative to the benign population.

At threshold 0.50, the test set instead had 1 false negative and 0 false positives, demonstrating the threshold trade-off.

9. Clustering: Diagnostic Pattern Segmentation

StandardScaler was applied before clustering. K-means was evaluated for K=2–6. K=2 had the highest silhouette score (0.343), while K=3 was selected for the requested multi-cluster profile reporting and produced interpretable segmentation. Hierarchical clustering with Ward linkage was used as a cross-check.



Cluster 1: n=110; M=110; B=0; M mix=100.0%; dominant standardized features: worst_area, worst_radius, mean_area, worst_perimeter, mean_perimeter.

Cluster 2: n=359; M=38; B=321; M mix=10.6%; dominant standardized features: mean_concavity, worst_concave_points, mean_concave_points, mean_compactness, worst_concavity.

Cluster 3: n=100; M=64; B=36; M mix=64.0%; dominant standardized features: worst_fractal_dimension, mean_fractal_dimension, worst_compactness, se_compactness, worst_concavity.

These cluster descriptions are non-clinical feature-pattern summaries and should not be interpreted as medical subtypes.

10. Recommendations, Limitations and Future Work

Recommendations: preserve the split discipline and seed; report ROC-AUC plus malignant recall, precision and F1; keep threshold selection separate from final test evaluation; use cluster profiles as communication aids.

Limitations: 569 observations, historical/specific dataset setting, uncertain generalization to other populations or institutions, and absence of external clinical factors. The project brief prohibits presenting the outputs as medical advice or a clinical diagnostic tool.

Future work: expand data, validate calibration, monitor drift, test threshold stability on new validation data, and consider human-in-the-loop review and explainability.

Appendix — Final Feature List

mean_radius, mean_texture, mean_perimeter, mean_area, mean_smoothness, mean_compactness, mean_concavity,
mean_concave_points, mean_symmetry, mean_fractal_dimension, se_radius, se_texture, se_perimeter, se_area, se_smoothness,
se_compactness, se_concavity, se_concave_points, se_symmetry, se_fractal_dimension, worst_radius, worst_texture, worst_perimeter,
worst_area, worst_smoothness, worst_compactness, worst_concavity, worst_concave_points, worst_symmetry, worst_fractal_dimension